

H CI Width vs. Number of Inferences

This section presents CI width versus number of inferences for all three models across varying templates and majority-vote sizes (K). Each figure corresponds to a specific model and K , with each column representing a different template. Within each subplot, the upper panel shows CI width vs. number of inferences, and the lower panel shows sorted CIs over 100 independent sample sets at the evaluation size of the bottom-right Pareto-frontier dot. Points with coverage in the range $99 \pm 1\%$ are shown as solid markers, while points with coverage $\geq 95\%$ are shown as semi-transparent markers. Efficiency gains are computed only for results achieving $99 \pm 1\%$ coverage.

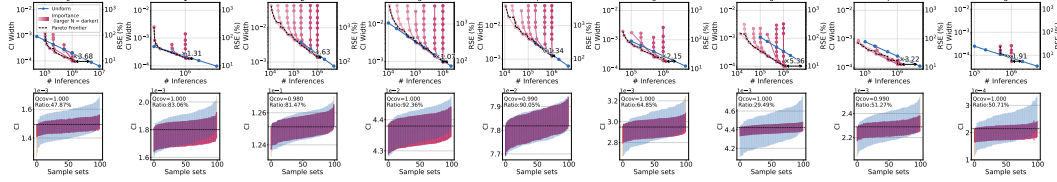


Figure 7: Qwen2.5-Math-7B-Instruct, $K = 4$, $\lambda = 0.1$

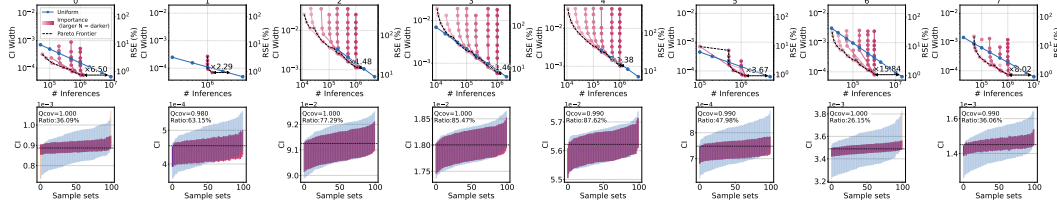


Figure 8: Qwen2.5-Math-7B-Instruct, $K = 8$, $\lambda = 0.1$

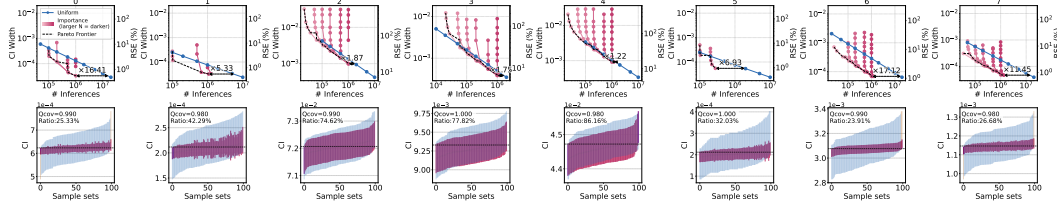
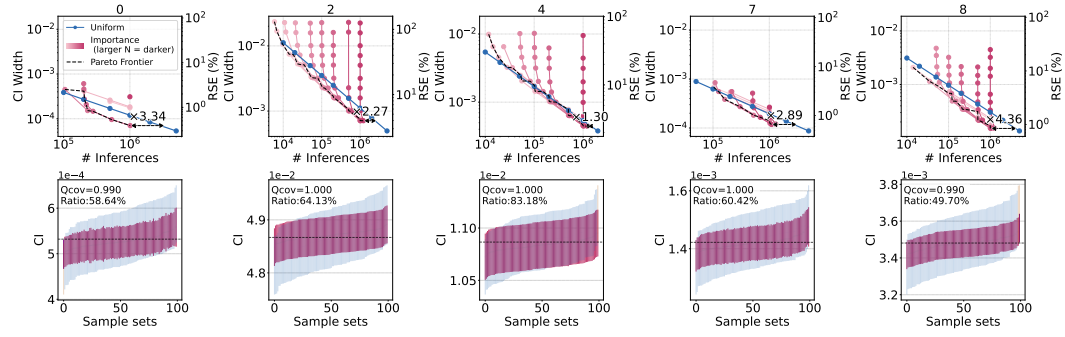
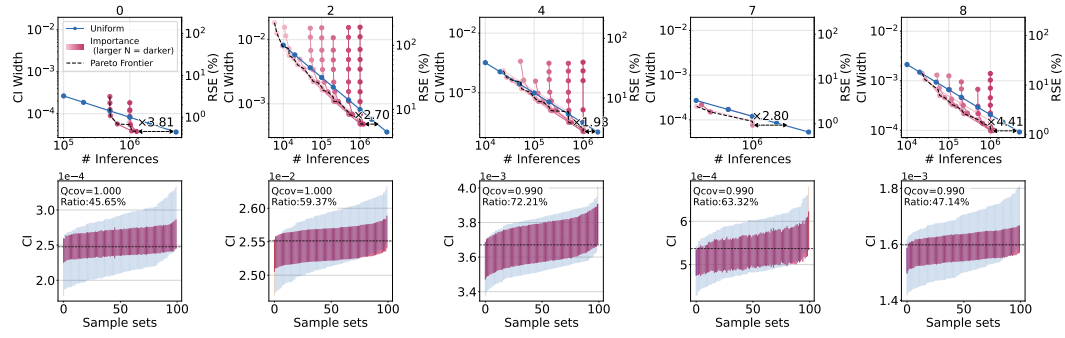
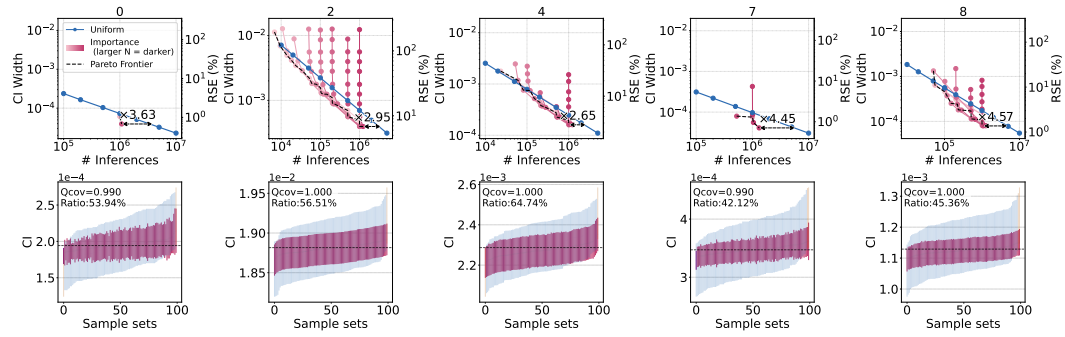
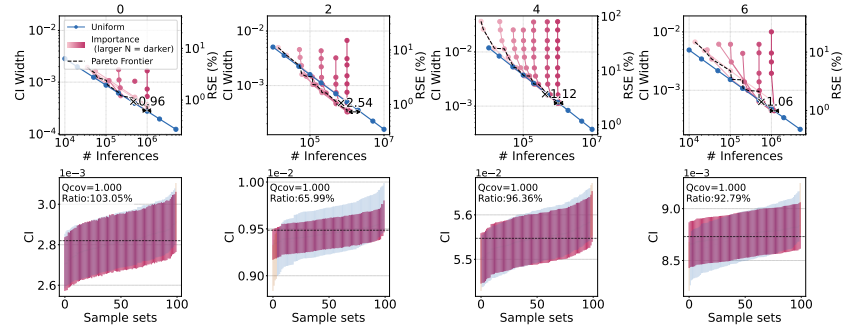
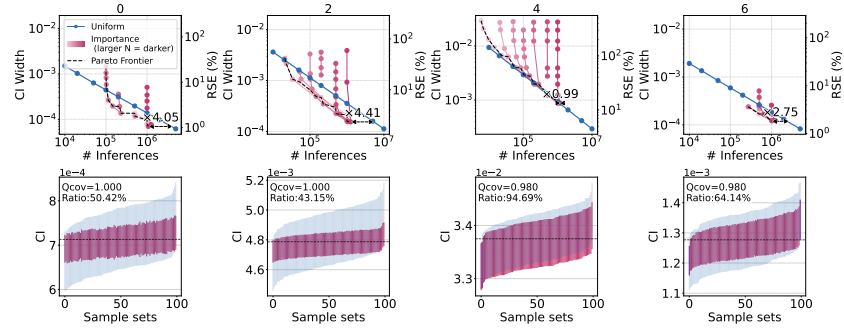
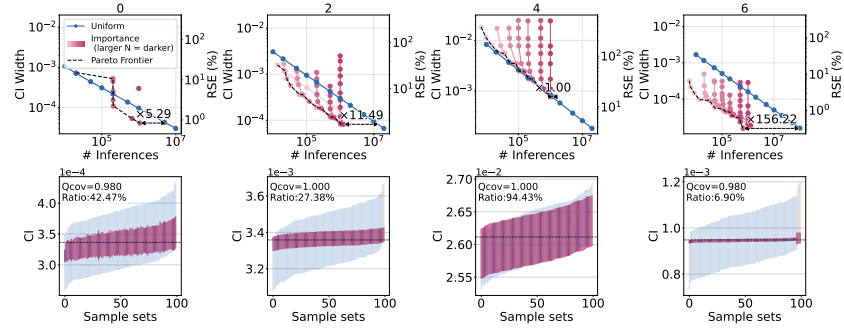


Figure 9: Qwen2.5-Math-7B-Instruct, $K = 16$, $\lambda = 0.1$

Figure 10: gpt-oss-20b-low, $K=8$, $\lambda=0.1$ Figure 11: gpt-oss-20b-low, $K=16$, $\lambda=0.1$ Figure 12: gpt-oss-20b-low, $K=24$, $\lambda=0.1$

Figure 13: Gemini 2.5 Flash Lite, $K = 4$, $\lambda = 0.1$ Figure 14: Gemini 2.5 Flash Lite, $K = 8$, $\lambda = 0.1$ Figure 15: Gemini 2.5 Flash Lite, $K = 16$, $\lambda = 0.1$